# LECTURE 6: SEARCH ALGORITHMS EXERCISES

DS 8001

# Ex1: Uninformed Search

Consider a search tree in which every node has $b = 3$ successors and the maximum depth is $m = 6$. The shallowest goal is at depth $d = 3$ and is the last node generated at that depth (worst case); other goals may exist deeper in the tree. The goal test is applied when a node is generated. When counting generated nodes, do not count the root.

(a) How many nodes does BFS generate, up to and including the goal? BFS stores every generated node (with its back pointer) and the root: about how many nodes are in memory when the goal is found?

(b) Iterative deepening runs depth-limited DFS with limits 1, 2 and 3. How many nodes does it generate in total? Why is this only a constant factor more than BFS?

(c) DFS (without a depth limit) keeps only the nodes on its stack and discards a subtree once it has been fully explored. At most how many nodes are on its stack? Is DFS guaranteed to return the shallowest goal?

(d) Memory is limited to 20 nodes, and the shallowest goal is required. Which of BFS, DFS and iterative deepening would you use?

# Ex2: Informed Search

Consider the search problem below, with start state $S$ and goal state $G$. The entry in row $X$ and column $Y$ is the cost of the directed edge $X \to Y$ ("–" means there is no edge), and the last column gives the heuristic value $h(X)$.

| from \ to | $S$ | $A$ | $B$ | $C$ | $D$ | $E$ | $G$ | $h$ |
|---|---|---|---|---|---|---|---|---|
| $S$ | – | 2 | 1 | – | – | – | 9 | 6 |
| $A$ | – | – | – | 2 | 3 | – | – | 0 |
| $B$ | – | – | – | – | 2 | 4 | – | 6 |
| $C$ | – | – | – | – | – | – | 4 | 4 |
| $D$ | – | – | – | – | – | – | 4 | 1 |
| $E$ | – | – | – | – | – | – | – | 10 |
| $G$ | – | – | – | – | – | – | – | 0 |

For each search below, give the order in which states are taken out of the priority queue (expanded), the path returned and its cost.

(a) Uniform-cost search (priority $g$)

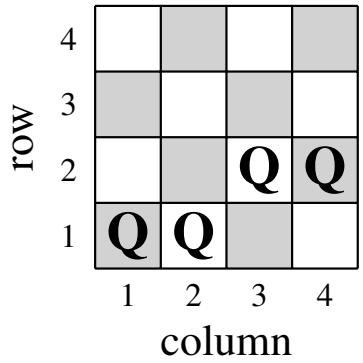
(b) Greedy best-first search (priority $h$)

(c) A search (priority $f = g + h$)

Conventions, as in the lecture's worked example: a state may enter the queue several times, once for each path that reaches it (tree search); the goal test is applied when a state is taken out of the queue; successors are added in alphabetical order, and ties are broken in favor of the entry added first.

# Ex3: Hill Climbing

Consider 4-queens with one queen per column: $s = (r_1, r_2, r_3, r_4)$, where $r_i$ is the row (1–4, bottom to top) of the queen in column $i$. Two queens attack each other if they share a row or a diagonal; $f(s)$ is the number of attacking pairs (minimize), and the number of conflicts of a queen is the number of queens it attacks. Use the neighborhood from the lecture: pick the most-conflicting column (the rightmost one, if tied) and move its queen to another row. Hill climbing moves to the best neighbor and stops when no neighbor is strictly better.




$s_0 = (1, 1, 2, 2)$

(a) Start from $s_0 = (1, 1, 2, 2)$. Compute $f(s_0)$ and the number of conflicts of each queen. Which column is chosen, and to which neighbor does hill climbing move?

(b) Continue until hill climbing stops. Check that $(2, 4, 1, 3)$ has $f = 0$. What kind of state has hill climbing stopped at?

(c) One run from a random start reaches $f = 0$ about 20% of the time. With independent random restarts, about how many runs (including the first) are needed on average? What is the probability that 10 runs all fail?

# Ex4: Simulated Annealing

Consider simulated annealing for a minimization problem. Let $s$ be the current state, $s'$ the neighbor picked as the candidate move, and $T(t)$ the temperature at time $t$. A move to a lower value, $f(s') < f(s)$, is always accepted; a move to a higher value is accepted with probability

$$p = \exp\left(-\frac{|f(s') - f(s)|}{T(t)}\right).$$

(a) If $f(s) = 6, f(s') = 5$ and $T(t) = 4$, and $s'$ has been picked as the candidate, what is the probability that we move from $s$ to $s'$? If instead the current state is $s'$ and $s$ has been picked, what is the probability that we move from $s'$ to $s$?

(b) If $f(s) = 0, f(s') = \ln 5$ and $T(t) = 1$, and $s'$ has been picked as the candidate, what is the probability that we move from $s$ to $s'$?

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# EX5: GENETIC ALGORITHM - 1

Each individual is a 5-bit string $x = (x_1, \ldots, x_5)$, and the reproduction probability of individual $s_i$ is fitness-proportional: $p_i = f(s_i)/\sum_j f(s_j)$. Consider the population

| $A$ | $B$ | $C$ | $D$ | $E$ |
|---|---|---|---|---|
| $(0,0,1,0,0)$ | $(0,1,0,0,1)$ | $(0,0,1,1,0)$ | $(0,0,0,1,0)$ | $(0,0,0,0,0)$ |

(a) Let $f(x) = \min\{t \in \{1,2,3,4,5,6\} : x_t = 1\}$, where $x_6 = 1$ is an extra dummy bit (so $f(x)$ is the position of the first 1 in $x$, and $f(x) = 6$ if $x$ contains no 1). Calculate the reproduction probabilities of $A, \ldots, E$.

(b) Let $f(x) = \max\{t \in \{0,1,2,3,4,5\} : x_t = 1\}$, where $x_0 = 1$ is an extra dummy bit (so $f(x)$ is the position of the last 1 in $x$, and $f(x) = 0$ if $x$ contains no 1). Calculate the reproduction probabilities of $A, \ldots, E$.

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# Ex6: Genetic Algorithm - 2

Consider the maximization problem

$$\max f(x) = -\frac{x^2}{10} + 3x, \qquad x \in \{0, 1, \ldots, 31\}.$$

Represent $x$ by its 5-bit binary code, from $x = 0 \equiv (00000)$ to $x = 31 \equiv (11111)$. Starting from an initial population of 10 random solutions, illustrate two generations (steps) of a genetic algorithm that works on these bit strings:

- select 10 parents with fitness-proportional (roulette-wheel) selection, $p_i = f(s_i)/\sum_j f(s_j)$; since $f(31) = -3.1 < 0$, use fitness 0 for $x = 31$;
- pair the parents in order (1st with 2nd, 3rd with 4th, ...) and apply one-point crossover at a random cut point;
- flip each bit of each child independently with a small mutation probability (e.g. 0.02);
- the 10 children replace the old population.

For each generation, show the population, the fitness values, the selection probabilities, and the results of crossover and mutation.